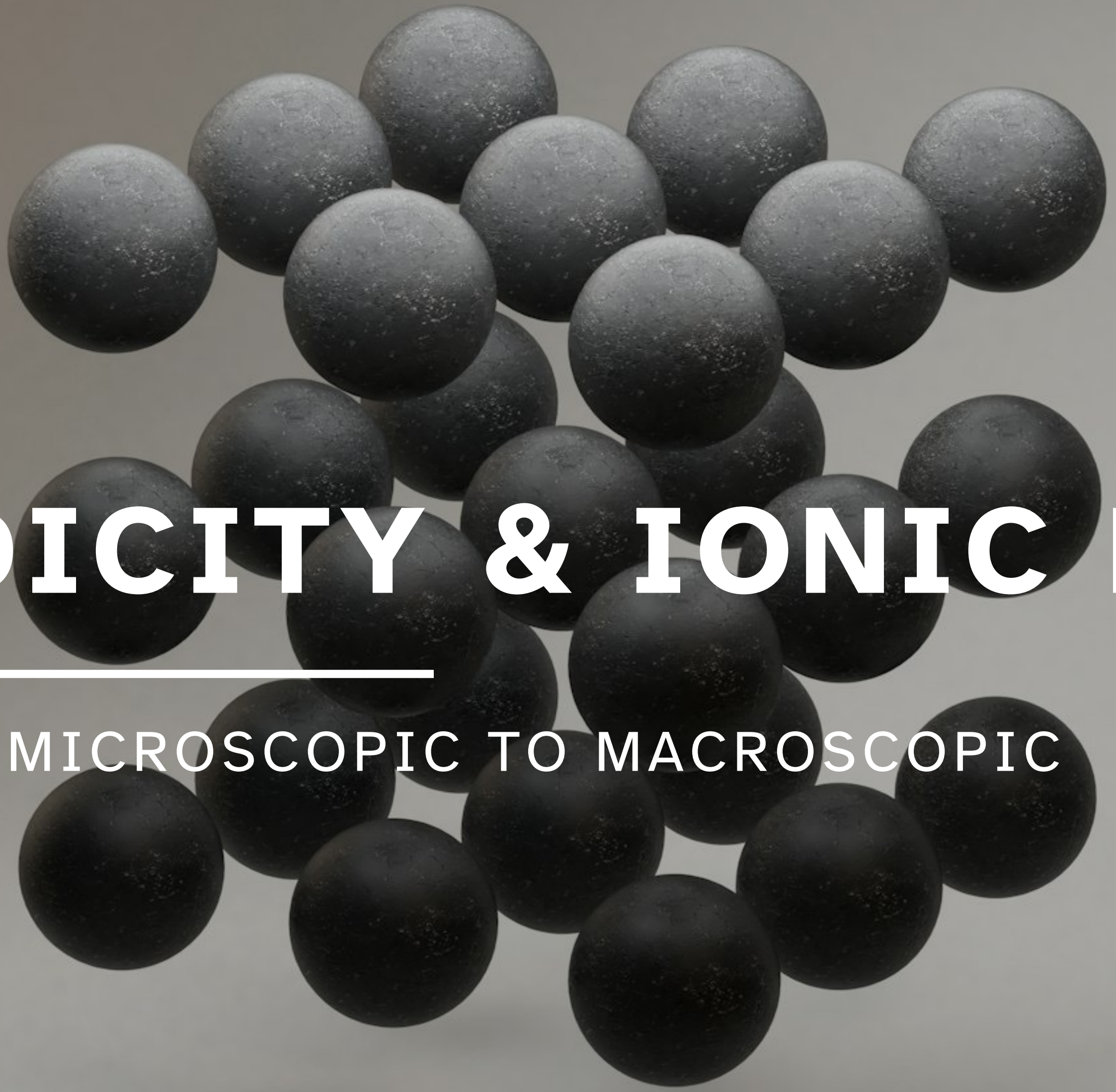


9B
PERIODICITY & IONIC BONDS

UNIT 04: FROM MICROSCOPIC TO MACROSCOPIC

CHEMISTRY 1310

MIOY T. HUYNH

LEARNING OBJECTIVES

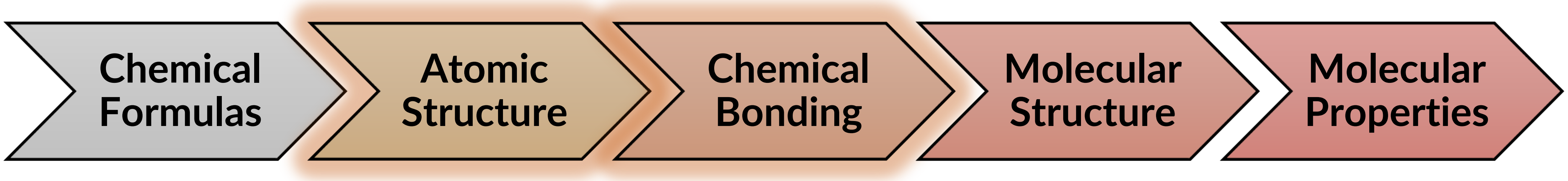
CHAPTER 9

1. Relate each electron configuration and position in the periodic table to the number of valence electrons for an element.
2. Apply Slater's rules to determine effective nuclear charge (Z_{eff}).
3. Predict and rationalize the periodic trends of atomic and ionic radius based on the concept of effective nuclear charge (Z_{eff}).
4. Predict the relative ionization energies of different elements from their positions in the periodic table.
5. Predict the relative electron affinities of different elements from their positions in the periodic table.
6. Determine the charges of common monatomic ions based on the positions of their elements on the periodic table and electron configuration.
7. Describe how electrostatic attractive forces allow ionic compounds to exist as very large, three-dimensional lattices.
8. Describe, apply, and rationalize the periodic trends in lattice energy.

Where did we start and where are we going?

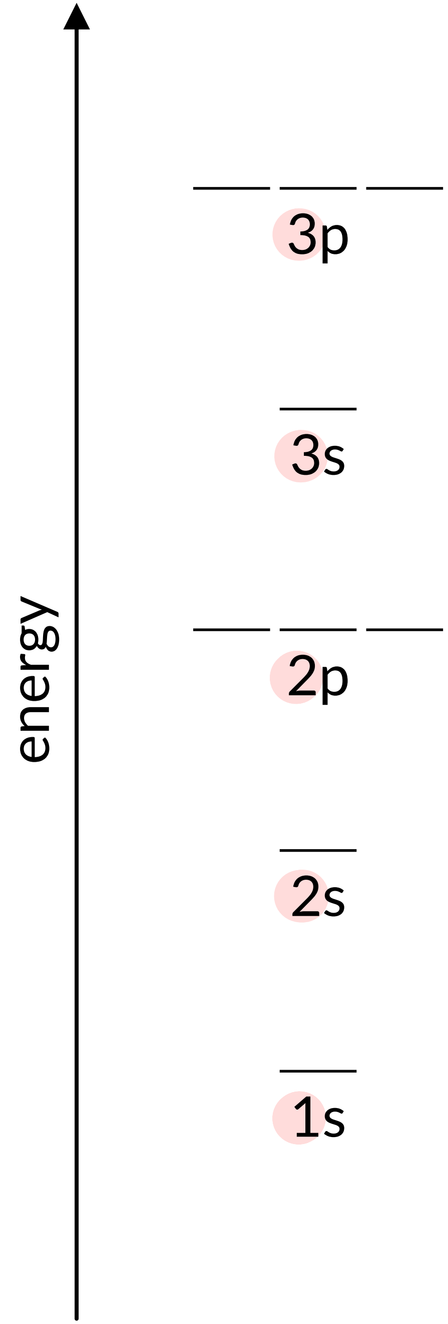
Chemistry in Context

What is our overall goal?



Ionization Energy and Electron Affinity

9.3



(higher n)
less energy
to remove as
e^- get farther
from nucleus

(lower n)
more energy
to remove as
e^- get closer
to nucleus

Relative periodic trends in ionization energy and atomic radius.

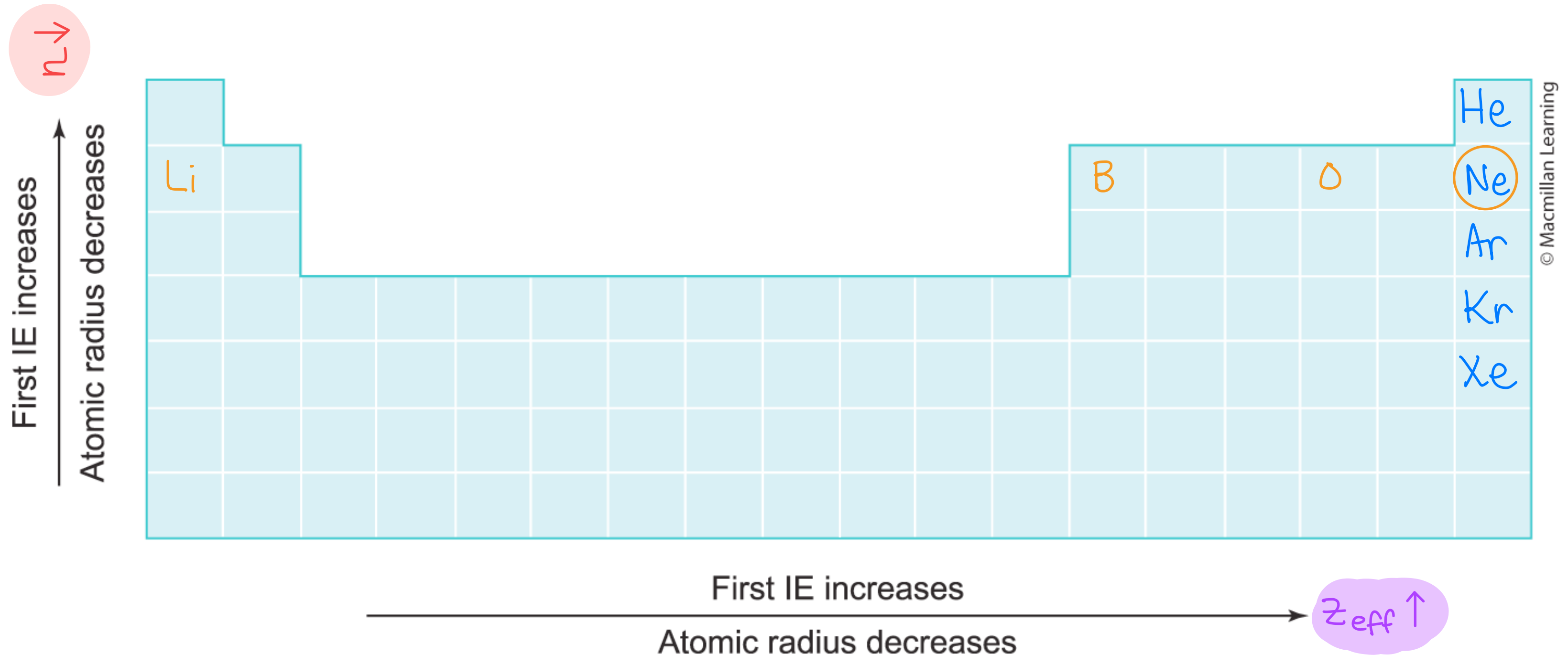


FIGURE 9.8

Ionization Energy and Electron Affinity

9.3

Ionization energy, IE

- The energy required to remove an electron from a neutral gaseous atom to produce a gaseous cation.
- Ionization energies (IE) are always positive.

$$Na(g) \rightarrow Na^{+}(g) + e^{-} \qquad IE_1 \qquad \text{(first ionization energy)}$$

neutral cation

remove this e^-

$\underbrace{1s^2 2s^2 2p^6}_{\text{core } e^-} 3s^1$

Ionization Energy and Electron Affinity

9.3

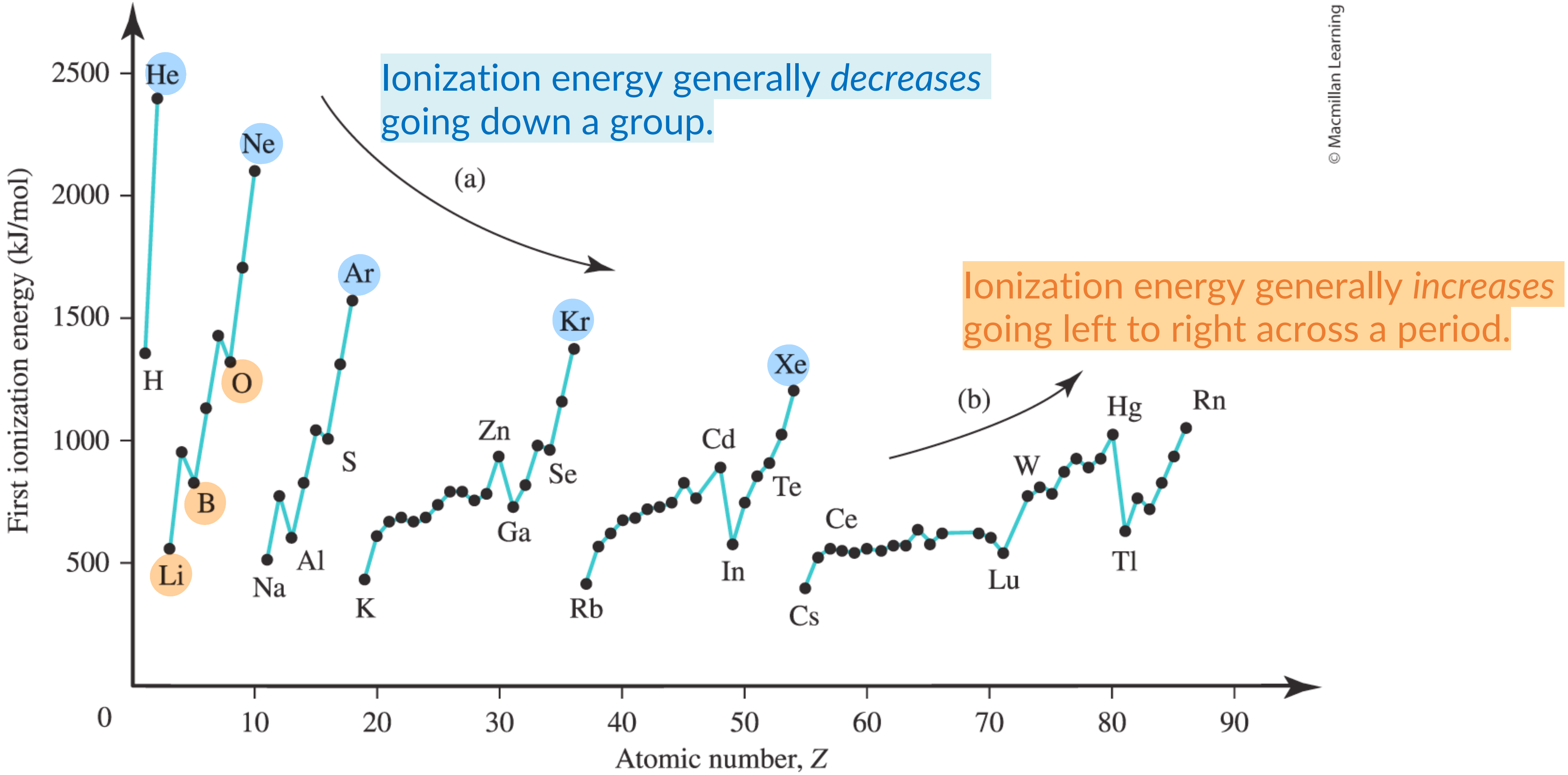


FIGURE 9.7

Ionization Energy and Electron Affinity

9.3

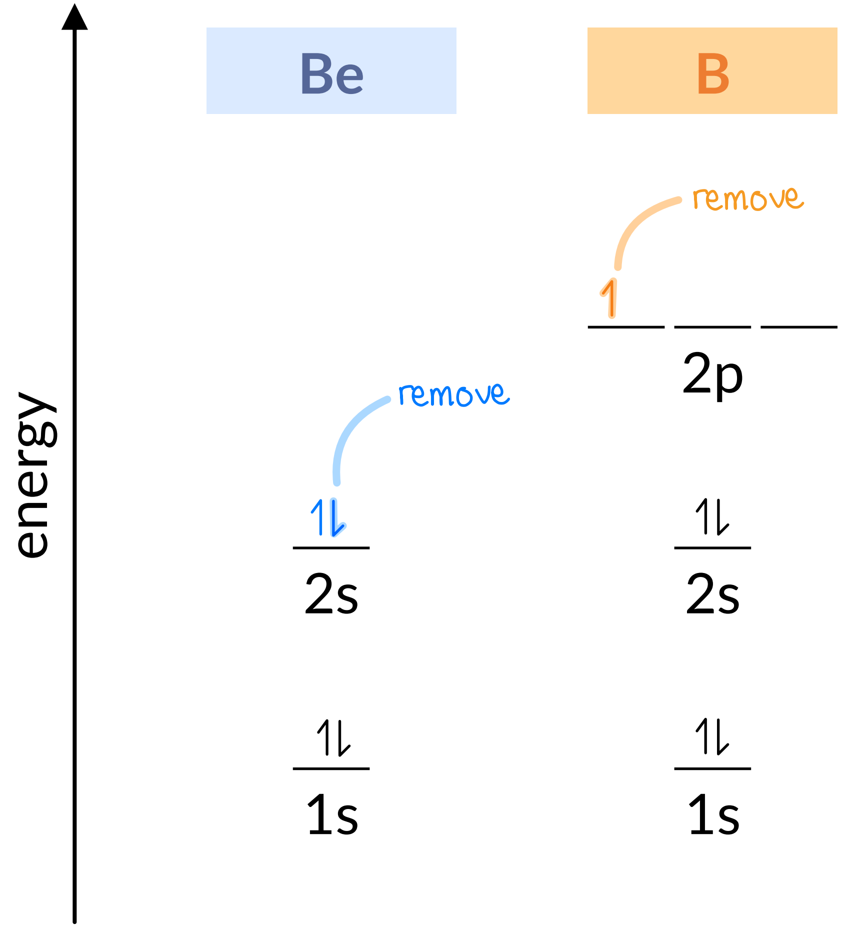
One exception to the general trend in ionization energy occurs when *new subshells* (*s, p, d, f*) begin to be occupied.

new subshells are → higher in energy,
farther from nucleus,
feel less nuclear charge,
→ easier to remove e^-
→ lower IE than expected

Example: Beryllium vs. Boron

$1s^2 2s^2$ $\quad$ $1s^2 2s^2 2p^1$

$IE_1(Be) > IE_1(B)$



Ionization Energy and Electron Affinity

9.3

Another exception to the general trend in ionization energy occurs when removing a *paired electron* from an orbital.

* a half-filled subshell tends to be fairly stable

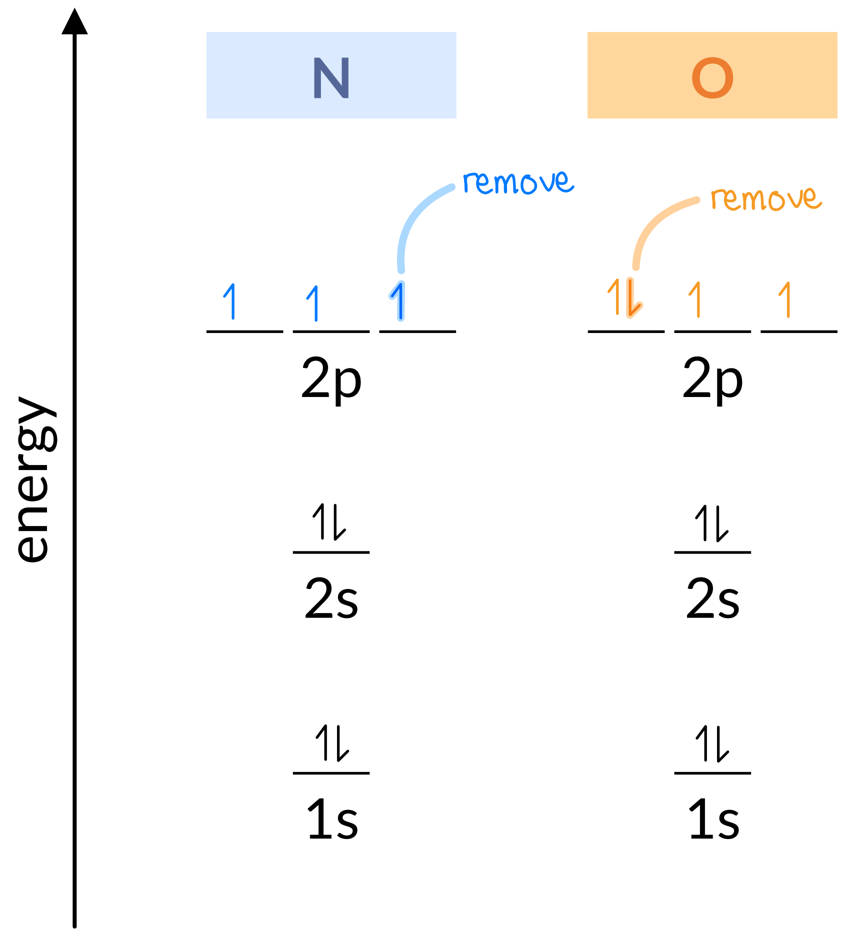
removing a paired e^- alleviates e^-/e^- repulsion $\longrightarrow$ leads to half-full subshell

$\longrightarrow$ easier to remove e^-

$\longrightarrow$ lower IE than expected

Example: Nitrogen vs. Oxygen $1s^2 2s^2 2p^3$ $1s^2 2s^2 2p^4$

$IE_1(N) > IE_1(O)$



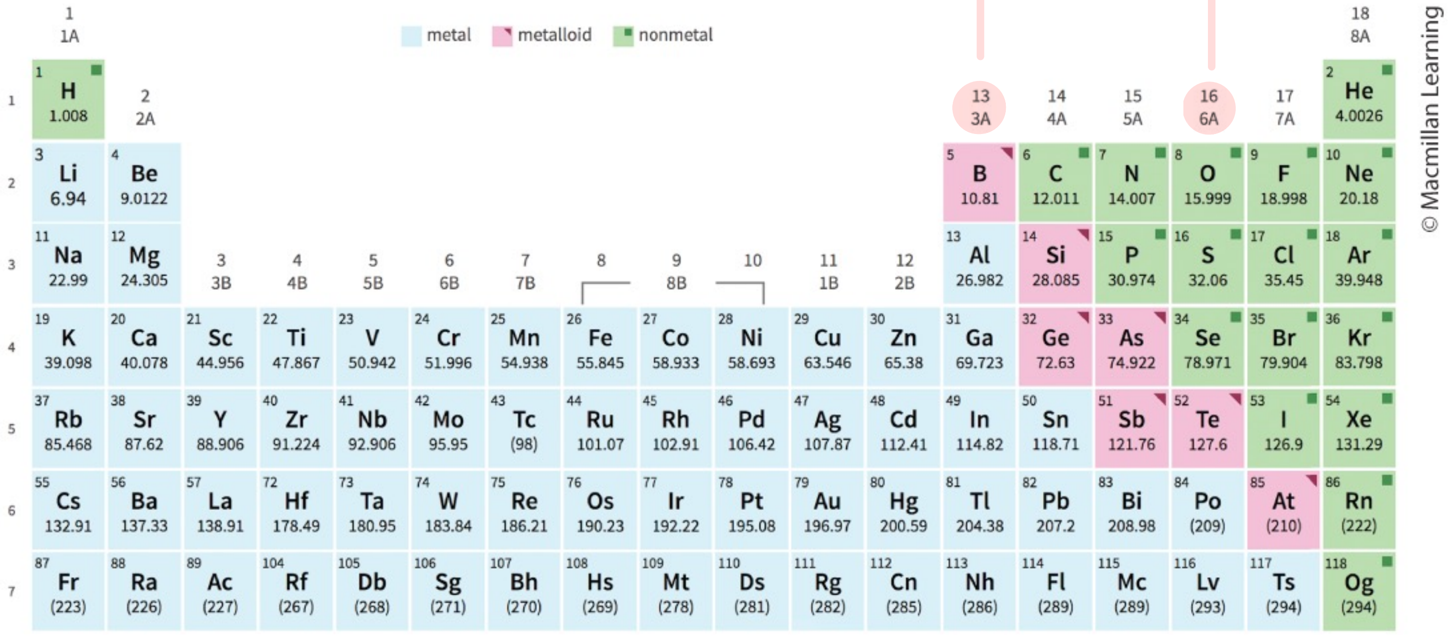
IN-CLASS QUESTION 1

Learning Objective(s): 9.4

Anomalies/Exceptions to periodic trends in first ionization energies occur for which two groups on the periodic table?

new subshell

leads to half-full subshell



Ionization Energy and Electron Affinity

9.3

Mg

Successive ionization energies (IE_n)

- For a given atom: $IE_1 < IE_2 < IE_3 < \ldots$
- First ionization energy (IE_1): removing the *first* e^- from a neutral atom.
- Second ionization energy (IE_2): removing the second e^-, this time from the +1 cation.
- Third ionization energy (IE_3): removing the third e^-, this time from the +2 cation.

$\uparrow\downarrow$ 3s

$\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow\downarrow$ 2p

$\uparrow\downarrow$ 2s

$\uparrow\downarrow$ 1s

$1s^2 2s^2 2p^6 3s^2$ valence

$1s^2 2s^2 2p^6 3s^1$

$1s^2 2s^2 2p^6$ core

$Mg(g) \longrightarrow Mg^+(g) + e^-$ IE_1

$Mg^+(g) \longrightarrow Mg^{2+}(g) + e^-$ IE_2

$Mg^{2+}(g) \longrightarrow Mg^{3+}(g) + e^-$ $IE_3 \gg IE_2 > IE_1$

$\boxed{\text{removing core } e^-\text{, smaller } n \text{ than valence, very costly!}}$

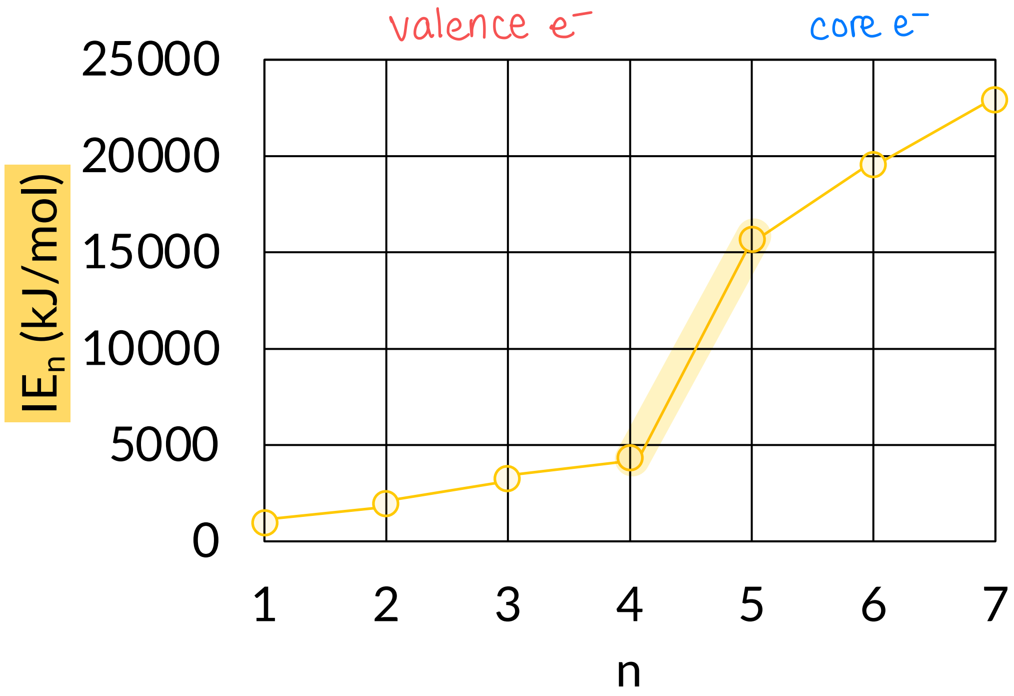
IN-CLASS QUESTION 2

Learning Objective(s): 9.4

Successive ionization energies (IE_n) are given for an unknown third-row element X. Based on this data, identify element X? *Hint: Write electron configurations and plot data!*

- ☐ Na
- ☐ Mg
- ☐ Al
- ☑ Si [Ne] $3s^2 3p^2$
- ☐ P
- ☐ S
- ☐ Cl
- ☐ Ar

first 4 e^-, $IE_1 \rightarrow IE_4$, must be valence e^-
fifth e^-, IE_5, must be first core e^-
→ element has 4 valence e^- → Si



TIP: look for huge jump!

IE_n	Value (kJ/mol)
IE_1	787
IE_2	1577
IE_3	3231
IE_4	4356
IE_5	16091
IE_6	19784
IE_7	23783

Ionization Energy and Electron Affinity

9.3

Electron affinity (EA)

- The change in energy (ΔE) when an electron is *added* to a neutral gaseous atom to form a gaseous anion.

$$e^- + Cl(g) \xrightarrow{EA} Cl^-(g)$$

* not reverse of ionization

EA + EA −

- Because it is a *change* in energy, EA can be positive *or* negative.
- A negative EA indicates that addition of an electron is ___favorable___.

- The trend in EA is not very consistent or obvious…

Ionization Energy and Electron Affinity

9.3

Generally, EA values of main group (*s*- and *p*-block) elements become *more negative* from left to right across a period, until reaching the noble gases where EA ≥ 0.

very unfavorable!

≥0 kJ/mol — −348.6 kJ/mol

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	H −72.8																	He ≥0
2	Li −59.6	Be ≥0											B −27.0	C −121.8	N ≥0	O −141.0	F −328.2	Ne ≥0
3	Na −52.9	Mg ≥0											Al −41.8	Si −134.1	P −72.0	S −200.4	Cl −348.6	Ar ≥0
4	K −48.4	Ca −2.4	Sc −18	Ti −8	V −51	Cr −65.2	Mn ≥0	Fe −15	Co −64.0	Ni −111.7	Cu −119.2	Zn ≥0	Ga −40	Ge −118.9	As −78	Se −195.0	Br −324.5	Kr ≥0
5	Rb −46.9	Sr −5.0	Y −30	Zr −41	Nb −86	Mo −72.1	Tc −60	Ru −101.0	Rh −110.3	Pd −54.2	Ag −125.9	Cd ≥0	In −39	Sn −107.3	Sb −101.1	Te −190.2	I −295.2	Xe ≥0
6	Cs −45.5	Ba −14.0	La −45	Hf ≥0	Ta −31	W −79	Re −20	Os −104.0	Ir −150.9	Pt −205.0	Au −222.7	Hg ≥0	Tl −37	Pb −35	Bi −90.9	Po −180	At −270	Rn ≥0

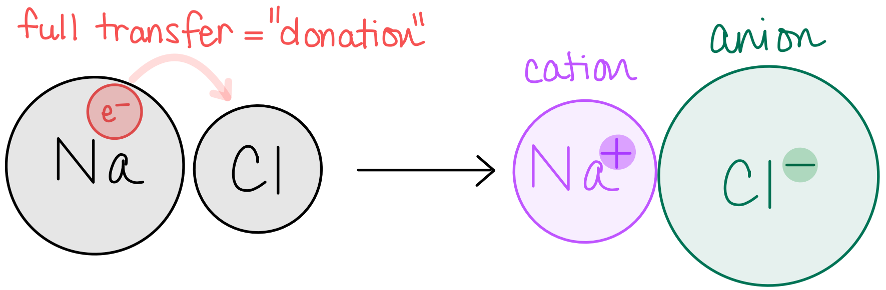
FIGURE 9.10

Ionic Bonding

9.4

Ionic “bond”

- Formed through the electrostatic attraction of two oppositely charged ions.



- *Example*: A neutral Na atom donates one electron to a neutral Cl atom to form the Na^+ cation and Cl^- anion that comprise the ionic compound NaCl.

Na $1s^2 2s^2 2p^6 3s^1$

Cl $1s^2 2s^2 2p^6 3s^2 3p^5$

same electron

$$Na \longrightarrow Na^+ + e^-$$

$$e^- + Cl \longrightarrow Cl^-$$

$$Na + Cl \longrightarrow NaCl$$

Na^+ $1s^2 2s^2 2p^6 = [Ne]$

Cl^- $1s^2 2s^2 2p^6 3s^2 3p^6 = [Ar]$

Ionic Bonding

9.4

Ionic lattice

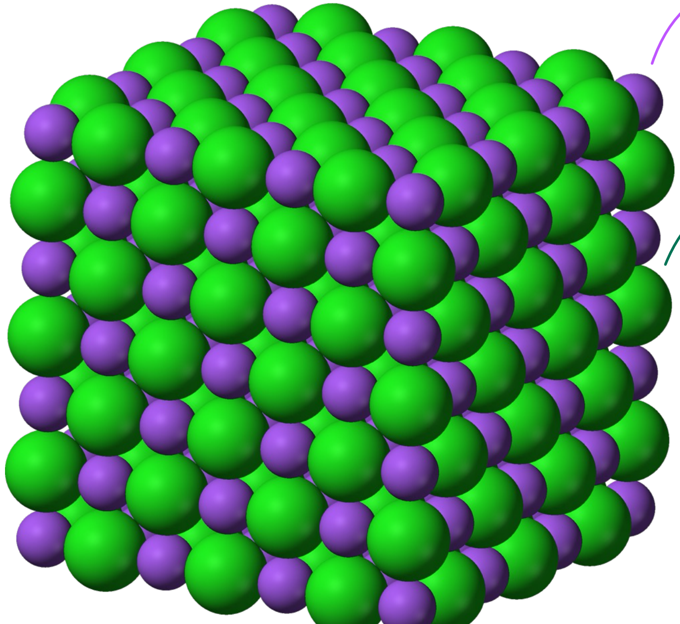
- Individual pairs of ions do not "bond" together.
- Rather, a large numbers of cations and anions form a 3D lattice.

Na^+

Cl^-

ionic compound represented as empirical formula

NaCl



space filling model

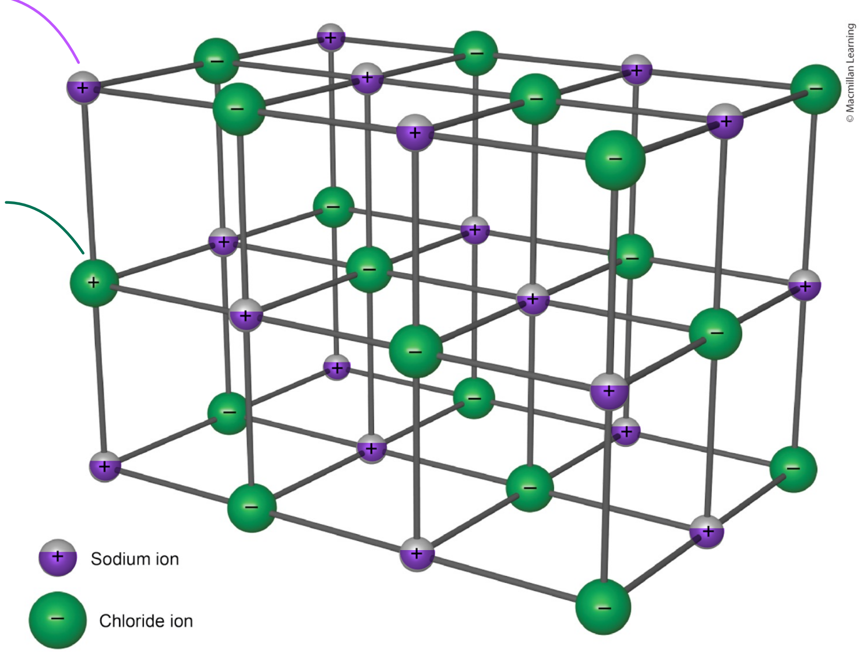


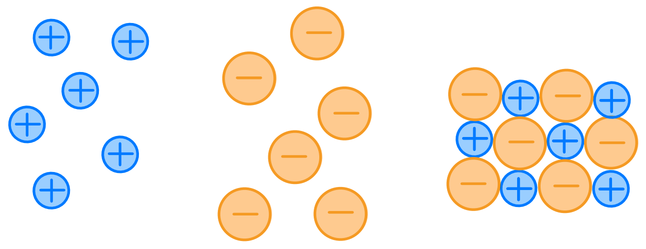
FIGURE 9.12

Lattice Energy

9.5

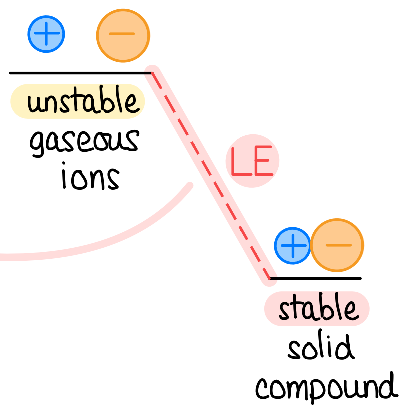
Lattice energy (LE)

- Accounts for contributions of all attractions and repulsions from ions.
- The lattice energy is the energy *released* when the gas-phase ions are converted into a neutral solid ionic compound.*



$K^+ (g) + Cl^- (g) \rightarrow KCl (s)$

unstable stable



negative

$LE = -717$ kJ/mol

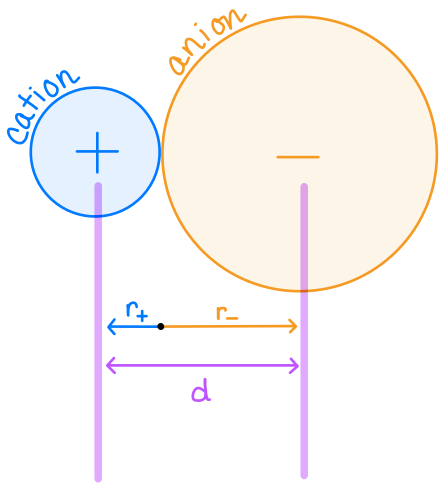
* Important Note: Some textbooks and resources define lattice energy as the energy that must be *absorbed* to break the lattice; the *reverse* process as written above. **To avoid confusion, we'll focus on the absolute value (or magnitude) when talking about "increasing" or "decreasing" lattice energy.**

Lattice Energy

9.5

The potential energy of the two interacting charged particles (E_{el}) reflects the *magnitude* of the lattice energy, $|LE|$.

Both LE and E_{el} tend to be *largest* in magnitude when: ions are small and highly charged



charge of one cation

charge of one anion

$$|LE| \propto E_{el} \propto \frac{q_+ \cdot q_-}{d}$$

$d = r_+ + r_-$

distance between ions = cation radius + anion radius

- E_{el} and $|LE|$ increases as charges (q_+ and q_-) increase
- E_{el} and $|LE|$ increases as ion distance ($d = r_+ + r_-$) decreases

Lattice Energy

9.5

- Smaller ions lead to stronger electrostatic attraction, resulting in lattice energies of larger magnitudes.

- Larger ions lead to weaker electrostatic attractions, resulting in lattice energies of smaller magnitudes.

FIGURE 9.15

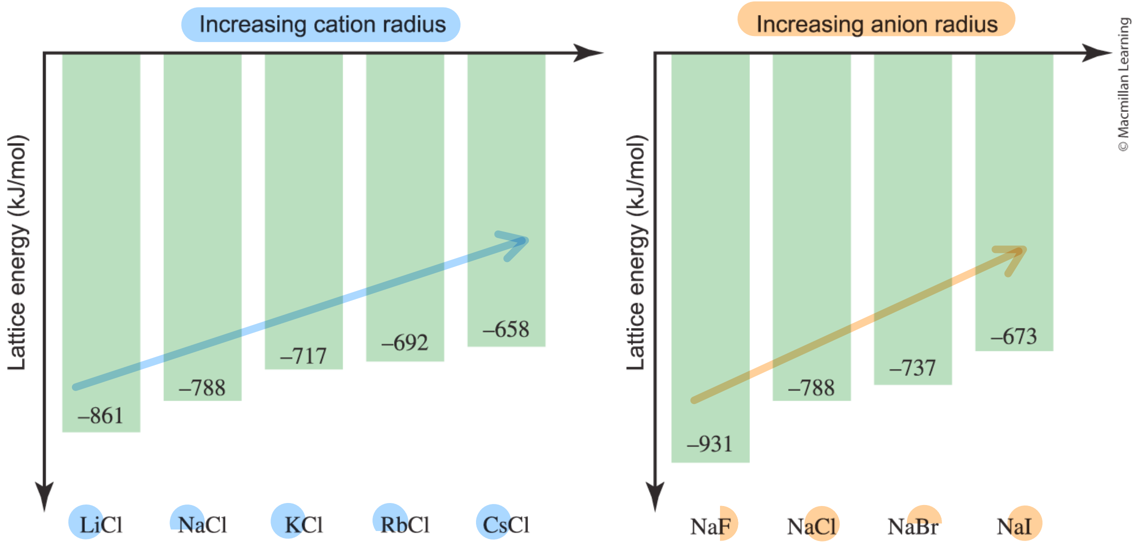
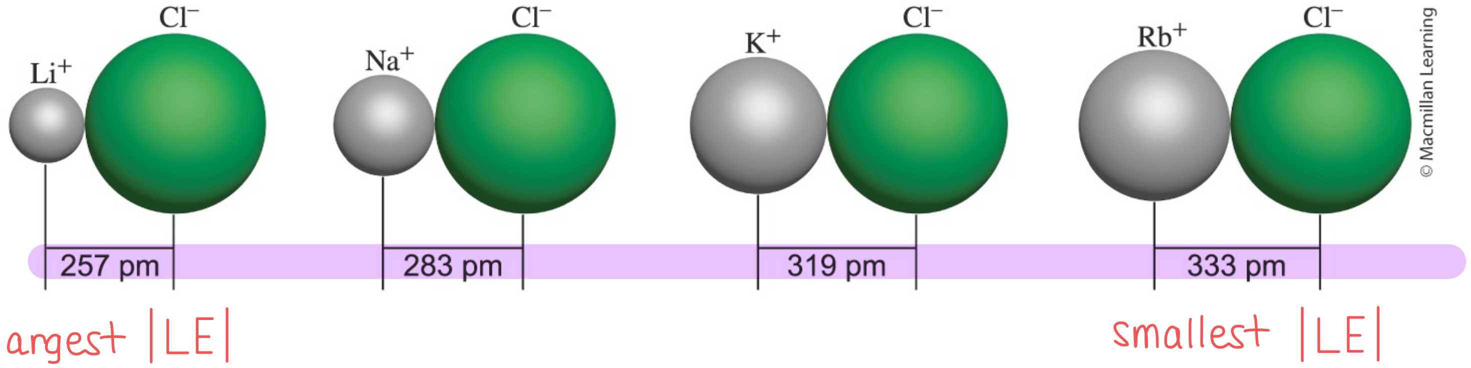


FIGURE 9.16

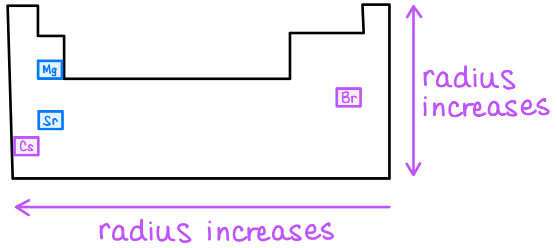


$d = r_+ + r_-$

EXAMPLE PROBLEM

Learning Objective(s): 9.6, 9.8

Considering the magnitude of lattice energies, order the following ionic compounds from largest to smallest lattice energy.



	cation	anion	check charges first	then distance/size
CsBr	Cs^{+}	Br^{-}	smallest charges ;	largest ions
SrO	Sr^{2+}	O^{2-}	2^{nd} largest charges ;	Sr^{2+}
CrN	Cr^{3+}	N^{3-}	largest charges	Mg^{2+} smaller than Sr^{2+}
MgO	Mg^{2+}	O^{2-}	2^{nd} largest charges ;	Mg^{2+}
Na_2S	Na^{+}	S^{2-}		

charge of one cation

charge of one anion

$$|LE| \propto E_{el} \propto \frac{q_+ \cdot q_-}{d}$$

$d = r_+ + r_-$

distance between ions

$CrN > MgO > SrO > Na_2S > CsBr$

BONUS QUESTION

Assessed on: Fall 2021, Exam 3

Which ionic compound is expected to have the *smallest value (in magnitude)* of lattice energy? In other words, the *least negative lattice energy*.

- ❑ LiF
- ❑ NaF
- ❑ KF
- ❑ RbF
- ❑ CsF